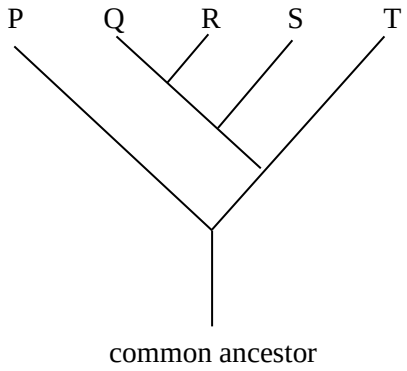
Multiple-choice questions

|!|BCEMB0300000001|!| ★

The following evolutionary tree shows the relationship between fives species, P to T.



Which pair of species are the most closely related?

A. P and Q
B. Q and R
C. R and S
D. S and T

□

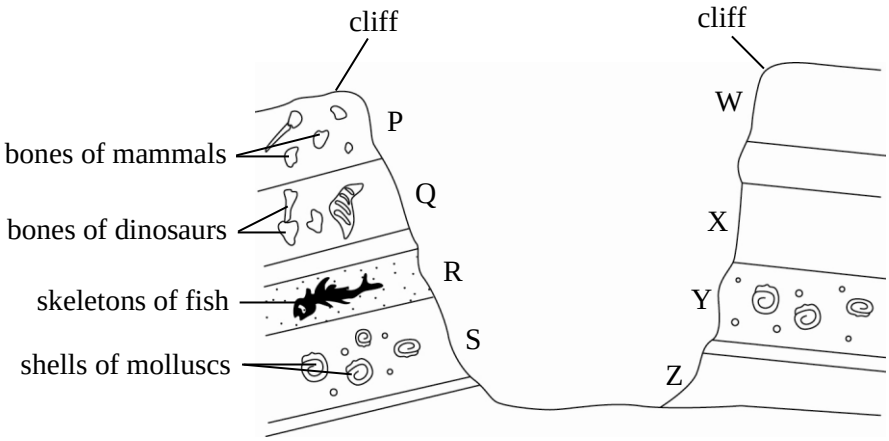
##B##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMA0300000002|!| ★★

Directions: The following two questions refer to the diagram below, which shows the cross section of rock layers in a valley.



Which layers of rock are most probably to have formed around the same period of time?

A. P and W
B. Q and X
C. R and Z
D. S and Y

☐

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0300000003|!| ★★

Which of the following statements about the rock layers in the diagram above is correct?

A. Fossils cannot be found in the surface layers of rock.
B. Older fossils are usually found in deeper layers of rock.
C. The deeper the rock layer, the more different types of fossil can be found.
D. Fossils cannot form in the bottom layer of rock.

☐

##B##

Skill assessed	Mark
Applying concepts	1

© 2020 Aristo Educational Press Ltd.

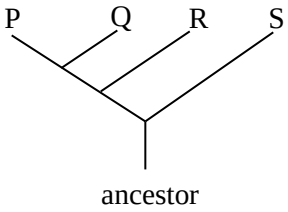
|!|BCEMB0300000004|!| ★

A type of protein can be found in species P, Q, R and S. The following table shows the number of amino acid differences between the protein found in species P and those in the other species.

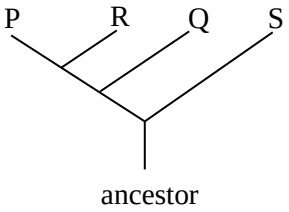
Species	Number of amino acid differences
Q	28
R	5
S	1

Which of the following diagrams ***best*** represents the evolutionary tree of the four species?

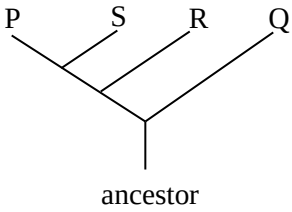
A.



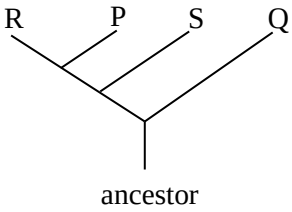
B.



C.



D.



##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0300000005|!| ★

It is suggested that reptiles might have evolved from amphibians. Which of the following characters might have gained by the reptiles that have evolved from amphibians?

A. constant body temperature
B. breathing with lungs
C. internal fertilization
D. carnivorous diet

##C##

Skill assessed	Mark
Applying concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMC0300000006|!|

Which of the following substances deposited in skeletons and is hardened to become fossils?

A. vitamins
B. mineral salts
C. carbohydrates
D. proteins

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0300000007|!|

Which of the following can be evidence for evolution of organisms?

(1) the fossil record
(2) comparison of DNA base sequence between organisms
(3) comparison of anatomy between organisms

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0300000008|!|

Which of the following can be determined from studying the fossil record?

(1) evolutionary history of different species
(2) time of existence of a species
(3) presence of genetic variation within a population

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

© 2020 Aristo Educational Press Ltd.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0300000009|!|

The Miller-Urey experiment in 1953 showed that

A. life could change into other forms.
B. life on Earth began from a common ancestor.
C. organic molecules could be broken down into simple inorganic molecules.
D. simple inorganic molecules could react to form organic molecules.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0300000010|!|

Which of the following statements about fossils is/are correct?

(1) Fossils are most commonly found in sedimentary rocks.
(2) Most fossils are formed from the bones and shells of organisms.
(3) Fossils present in the upper rock layers are earlier than those present in lower rock layers.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0300000011|!| ★

Which of the following ways can help scientists estimate the age of fossils?

(1) comparing the quantity of ^{14}C in fossils
(2) measuring the hardness of the fossils
(3) comparing the quantity of ^{40}K in the rocks near the fossils

© 2020 Aristo Educational Press Ltd.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

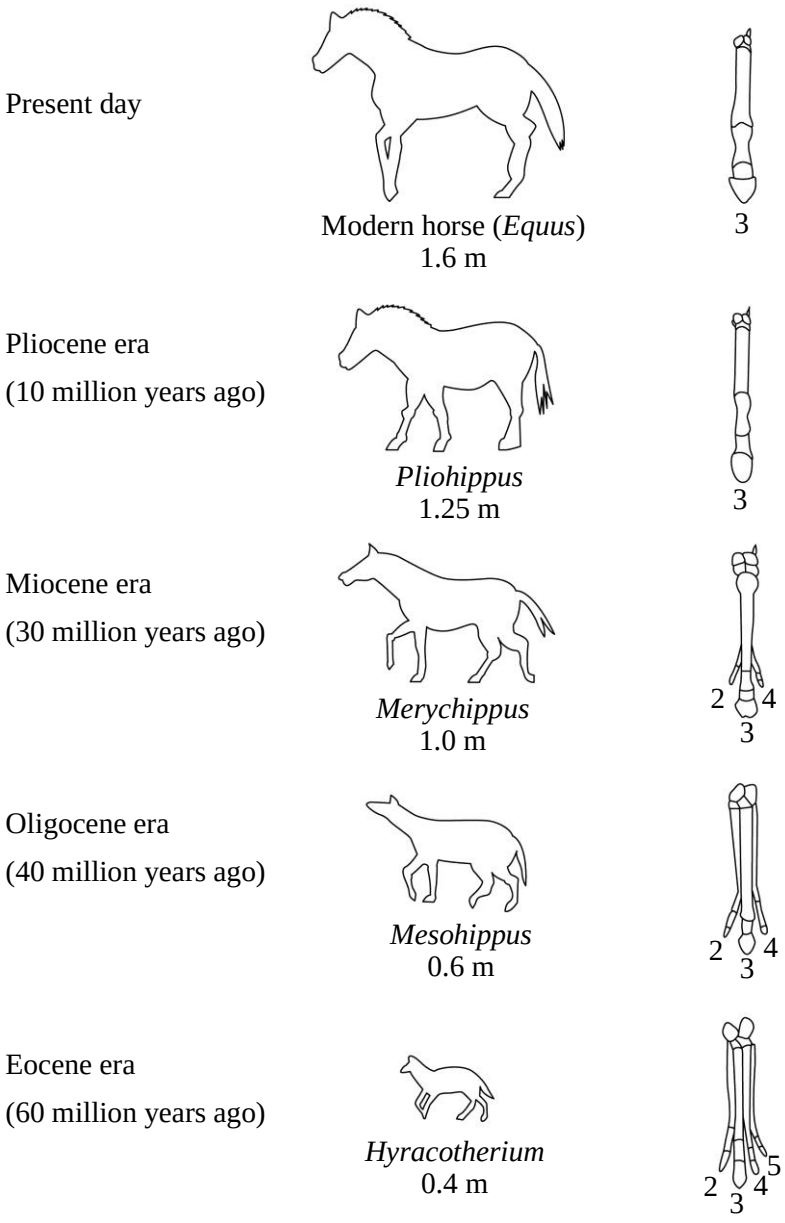
□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0300000012|!| ★

Directions: The following two questions refer to the diagram below, which shows some of the changes in horses that occurred over the past 60 million years. The different toes are labelled with numbers. All the horses are extinct except the modern horse.



© 2020 Aristo Educational Press Ltd.

Which of the following statements about the horses is correct?

A. *Pliohippus* was four times as tall as *Hyracotherium*.
B. *Mesohippus* had the same number of toes as *Hyracotherium*.
C. The fifth toe of horse was lost in the Miocene era.
D. The greatest reduction in the size of the second and fourth toes of horse occurred in the Oligocene era.

☐

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0300000013|!| ★

The changes in horse as shown in the diagram provide evidence which suggests

A. domestication.
B. that horses gradually became more complex.
C. that the characteristics of horses changed over time.
D. the modern horse would not evolve.

☐

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0300000014|!| ★★

Which of the following are homologous features?

A. fin in fish and wings in birds
B. the striped coat of the zebra and the tiger
C. wings in bats and bees
D. human arms and wings in bats

☐

##D##

Skill assessed	Mark
Applying concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0300000015|!| ★

Which of the following statements about homologous structures is correct?

A. They have the same ancestral origin and always serve the same function.
B. They have different ancestral origins but always serve the same function.
C. They have the same ancestral origin but may have adapted to serve different functions.
D. They have different ancestral origins and may have adapted to serve different functions.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0300000016|!| ★

Fossils of the earliest life forms on Earth are seldom discovered because

A. the earliest life forms have all evolved into more complex species.
B. the earliest life forms have all become extinct before they could become fossilized.
C. the soft bodies of the earliest life forms did not fossilize well.
D. all of the above.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0300000017|!| ★

Directions: The following three questions refer to the table below, which shows the taxonomic groups of rabbits and hares which are closely related species.

	Rabbits	**Hares**
Kingdom	Animalia	Animalia
Phylum	Chordata	Chordata
Class	Mammalia	Mammalia
Order	Lagomorpha	Lagomorpha
Family	Leporidae	Leporidae
Genus	*Sylvilagus* and 7 other genera	*Lepus*

© 2020 Aristo Educational Press Ltd.

Which of the following taxonomic groups shows that rabbits and hares possess vertebrae?

A. kingdom
B. phylum
C. class
D. order

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0300000018|!| ★

Which of the following ***cannot*** provide evidence to show that rabbits and hares have evolved from a common ancestor?

A. similarities in their embryos
B. similarities in their skeletal structures
C. similarities in the amino acid sequences of proteins
D. similarities in the number of types of DNA bases present

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0300000019|!| ★

Rabbits and hares are both herbivores and have teeth adapted for this diet.
Which of the following is ***not*** a possible factor for the development of the specialized teeth in rabbits and hares?

A. evolution of predators
B. changes in the environment
C. variations within rabbits and hares
D. competition among rabbits and hares

##A##

Skill assessed	Mark
Applying concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0300000020|!| ★

Which of the following is/are the limitation(s) of using the fossil record as evidence for evolution?

(1) Some organisms may not form fossils.
(2) Some organisms had been preserved in amber instead of sedimentary rocks.
(3) Some fossils are damaged and incomplete.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0300000021|!| ★

The table below shows the amino acid sequences of part of the insulin polypeptide chain of different organisms. (Each letter represents one type of amino acid.)

Organisms	Amino acid sequence																
Human	R	R	E	A	E	D	L	Q	V	G	Q	V	E	L	G	G	G
Guinea pig	R	R	E	L	E	D	P	Q	V	E	Q	T	E	L	G	M	G
Pig	R	R	E	A	E	N	P	Q	A	G	A	V	E	L	G	G	G
Cow	R	R	E	V	E	G	P	Q	V	G	A	L	E	L	A	G	G
House mouse	R	R	E	V	E	D	P	Q	V	E	Q	L	E	L	G	G	S

According to the table above, which of the following organisms has the closest evolutionary relationship with humans?

A. guinea pig
B. pig
C. cow
D. house mouse

##B##

Skill assessed	Mark
Analysing data	1

© 2020 Aristo Educational Press Ltd.

|!|BCEMB0300000022|!| ★

Directions: The following two questions refer to the information below.

By comparing the DNA base sequences of organisms, we can determine the evolutionary relationships among organisms. The following shows part of the DNA base sequences of four organisms.

Human:	CCA	TAG	CAC	CTA
Pig:	CCA	TGT	AAA	CGA
Chimpanzee:	CCA	TAA	CAC	CTA
Cricket:	CCT	AAA	GGG	ACG

Based on these sequences, which two organisms are ***most*** closely related?

A. human and pig
B. human and chimpanzee
C. pig and chimpanzee
D. pig and cricket

□

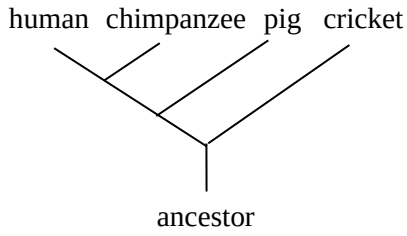
##B##

Skill assessed	Mark
Analysing data	1

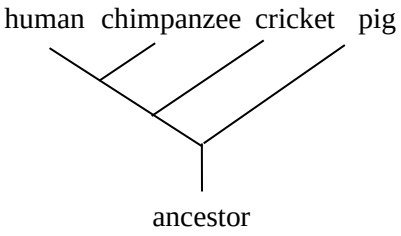
|!|BCEMC0300000023|!|

Which of the following diagrams ***best*** represents the evolutionary tree of the four species?

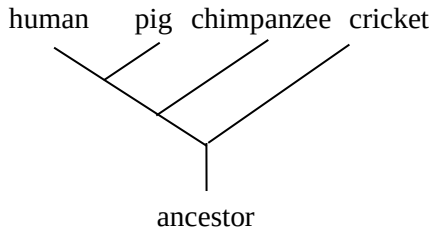
A.



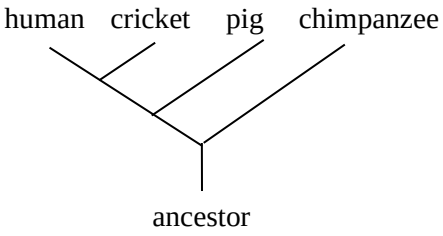
B.



C.



D.



□

##A##

© 2020 Aristo Educational Press Ltd.

Skill assessed	Mark
Analysing data	1

|!|BCEMB03000000024|!| ★

The table below shows part of the amino acid sequence in a polypeptide chain of a protein in four species.

Species	Amino acid sequence
W	serine - asparagine - histidine - leucine - tyrosine
X	serine - histidine - leucine - isoleucine - isoleucine
Y	serine - asparagine - histidine - leucine - tyrosine
Z	serine - histidine - asparagine - leucine - tyrosine

Which species is the least closely related to other species?

A. species W
B. species X
C. species Y
D. species Z

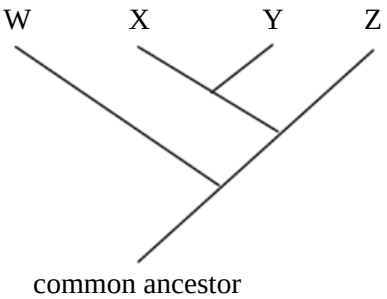
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB03000000025|!| ★

Directions: The following two questions refer to the evolutionary tree below.



Which two species are the most closely related?

A. W and X
B. X and Y
C. Y and Z
D. Z and W

□

© 2020 Aristo Educational Press Ltd.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB03000000026|!| ★

The four species are orangutan, gorilla, chimpanzee and humans. The percentage similarity in DNA base sequence between humans and

- orangutan is 96.9%.
- gorilla is 98.4%.
- chimpanzee is 98.8%.

Which of the following combinations correctly identifies X, Y and Z?

	X	***Y***	***Z***
A.	chimpanzee	humans	orangutan
B.	chimpanzee	gorilla	orangutan
C.	humans	chimpanzee	orangutan
D.	humans	chimpanzee	gorilla

□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0300000027|!| ★

Which of the following is the ***least*** accurate for deducing the evolutionary relationship between different species?

A. comparing external features of organisms
B. studying fossils of organisms
C. comparing the amino acid sequences of common proteins in different organisms
D. comparing the DNA base sequences of different organisms

□

##A##

Skill assessed	Mark
Applying concepts	1

© 2020 Aristo Educational Press Ltd.

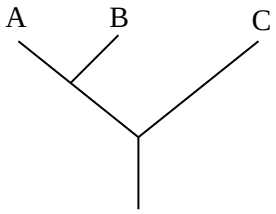
|!|BCEMB0300000028|!| ★

Haemoglobin is a protein found in most vertebrates. The amino acid sequences of haemoglobin in three species of mammals were analysed and compared. The following table shows the similarity in the amino acid sequences of haemoglobin among the three species:

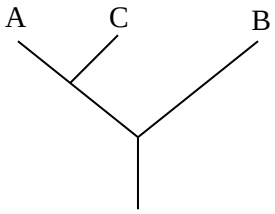
	Species B	**Species C**
Species A	71%	95%
Species B		72%

Assuming that the three species of mammals arose from a common ancestor, which of the following ***best*** represents the possible evolutionary relationships among the three species?

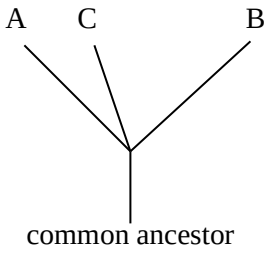
A.



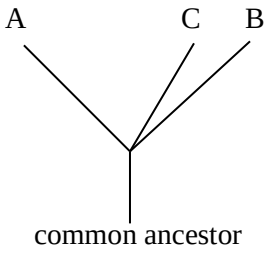
B.



C.



D.



□

##B##

Skill assessed	**Mark**
Analysing data	1

© 2020 Aristo Educational Press Ltd.